

Supplementary Material: Measuring Counseling Interaction Patterns Across Human–Human and Human–LLM Dialogues

Abstract

This document accompanies the main article. It reports the full persona list, complete client- and counselor-side OnCoCo label distributions, hidden-Markov state summaries and state-count sensitivity, transition heatmaps, sentence-embedding projections, per-conversation within-condition variability, the RQ3 multiplicity control, the noise-band placement under each unit of analysis, the classifier-confidence diagnostics, and the pre-declared category exclusion sets with their leave-one-out sweep. Tables and figures are numbered S1, S2, ... and are referenced by those numbers in the main text.

Keywords

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1. Prompts and Case Descriptions

The prompt templates that drove the simulated client are released in the companion repository (directory `prompts/`), one file per persona and template variant, with the conversation history that the platform inserts at generation time replaced by a placeholder. The repository also contains the HH roleplay and H–LLM conversations with personal data masked, the OnCoCo labels aligned to the masked text, the persona definitions, and the segmentation adapter. Prompts of the earliest deployment (Mixtral 8x7B) were not logged by the platform at that time. HH real conversations are not released because they stem from real help-seekers.

2. Persona List

3. Additional Tables and Figures

This appendix reports full label distributions, persistence statistics, within-condition variability, transition heatmaps, and embedding projections.

4. Pre-Declared Category Exclusion Sets

The sensitivity analysis reported in Section 5.6.1 of the main paper removes categories that are arguably fixed by the training setting rather than by counseling behavior. The first two sets below were declared before the analysis was run. S3 is post-hoc and is flagged as such. All three are reproduced here so the choice is auditable.

S1: setting-bound (primary). *Objective of the assignment* (CL-IF-AO-*Obj-*): and *Extension of the assignment* (CL-IF-AO-*Ext-*): the assignment is supplied by the persona prompt and the course task, so a simulated client states it because it is in its context window. *Definition of counseling objectives* (CO-IF-AO-*ICO-*): the counselor mirror of the same construct. *Formalities at the beginning* for both roles (CL-FB-*-*-*, CO-FA-*-*-*): the opening formalities are platform-dictated. Archived real conversations open with a system join event, and H–LLM sessions open with a scripted first turn. *Inappropriate remark* for both roles (CL-O-*-*-UCO-*, CO-O-*-*-UCO-*): excluded not as a setting artifact but as a *classifier* artifact. It carries the lowest median confidence of any label with appreciable support (median 0.48–0.60, against 0.88 or above for every category carrying the behavioral signature). The counselor-side code also absorbs 6.5% of HH real counselor spans against about 1% in the other two conditions. The client-side code is predicted with higher confidence (0.70–0.78) and is excluded only for symmetry with it. This was established in the confidence diagnostics before the present analysis was run.

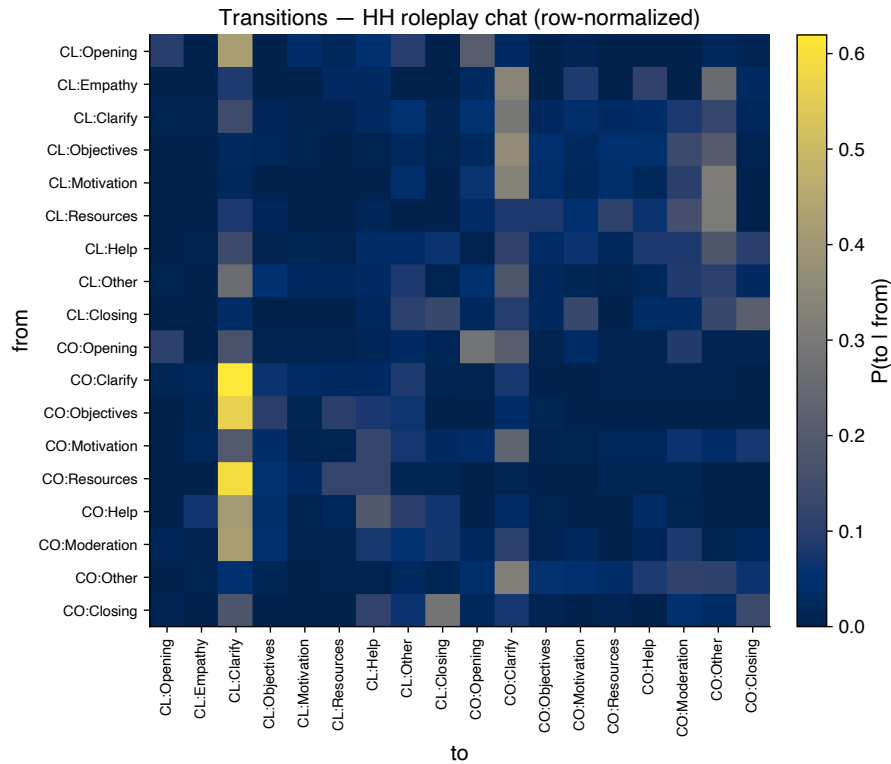


Figure S1. Transition heatmap for HH roleplay (row-normalized).

S2: measurement residual (secondary). *Other statements* for both roles (CL-O-***-O-*, CO-O-***-O-*), the residual category for units the classifier cannot place elsewhere. It is kept as a separate tier because it tests a different threat, namely measurement error rather than a setting confound. Merging the two would let one exclusion set carry two arguments.

S3: closing formalities (tertiary, reported but not pre-declared). *Farewell* for both roles (CO-FC-***-F-*, CL-FC-***-F-*). Session end is partly imposed by the time frame of the exercise, but closing is itself a counseling skill, so the case is weaker and we report rather than assume it.

Deliberately not excluded: Consent (CL-IF-ACP-**-Cons-*). It captures brief acknowledgements and agreement rather than a setting artifact. It is the second-largest client-side category and the largest single driver of the gap. Removing the largest contributor *because* it is large would be the post-hoc selection the objection warns against. It appears instead as one point among all others in the leave-one-out sweep (Figure S9), where removing it *increases* the gap.

5. Methodological Details Referenced from the Main Text

HMM initialisation. We fit each condition from 20 random initialisations and retain the best-likelihood solution, because the EM likelihood surface for these sequences is strongly multimodal. Across conditions, 20 random initialisations recover 6–11 distinct optima. A single fixed initialisation lands in a markedly inferior one for all three chat conditions (up to $\Delta \log L = 1,468$), and the resulting states then have near-duplicate output distributions. Table S6 reports the sensitivity sweep over $k = 2 \dots 8$, in which the cross-validated optimal state count tracks corpus size rather than any property of the conditions.

Misspecified segmentation. The bottom row of Table 3 in the main paper is a naive character-class split. It produces 16.2 fragments per message with a mean length of 10 characters, because it splits on a frequent letter rather than only on sentence-final punctuation. The classifier can assign the resulting fragments only to residual categories. Under such a split the between-condition distances collapse by a factor of three to four, and the client-side gap becomes the single unit-by-role combination that fails to reach significance. The row is included in the main paper so that the contrast with a correctly specified sentence split is visible. It is not a segmentation comparison.

Cross-course association (RQ3, secondary). Across all six courses with unambiguous survey linkage, client-side JSD and disagreement with the realism item correlate at Spearman $\rho = .83$ ($p = .042$). The result is fragile. Two of the six courses contribute only 2 and 9 respondents, and dropping them leaves the direction intact but not the significance ($\rho = .80$, $p = .20$ at $k = 4$). A Pearson specification on all six gives $r = .68$ ($p = .13$). We therefore report it as descriptive support for the within-course-line result of Section 5.7 in the main paper rather than as independent evidence.

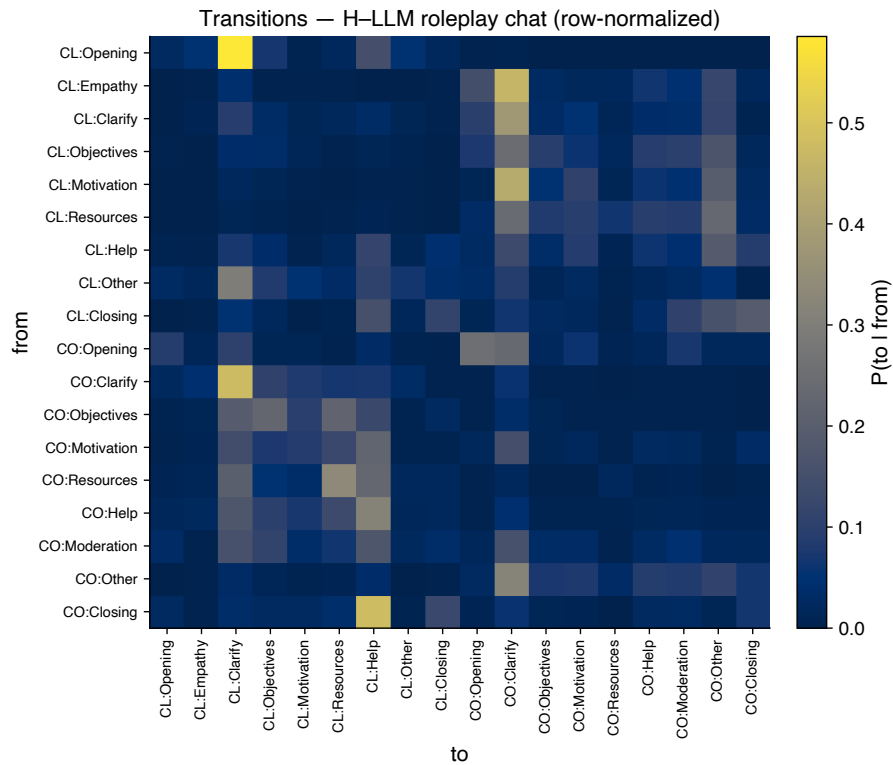


Figure S2. Transition heatmap for H-LLM roleplay (row-normalized).

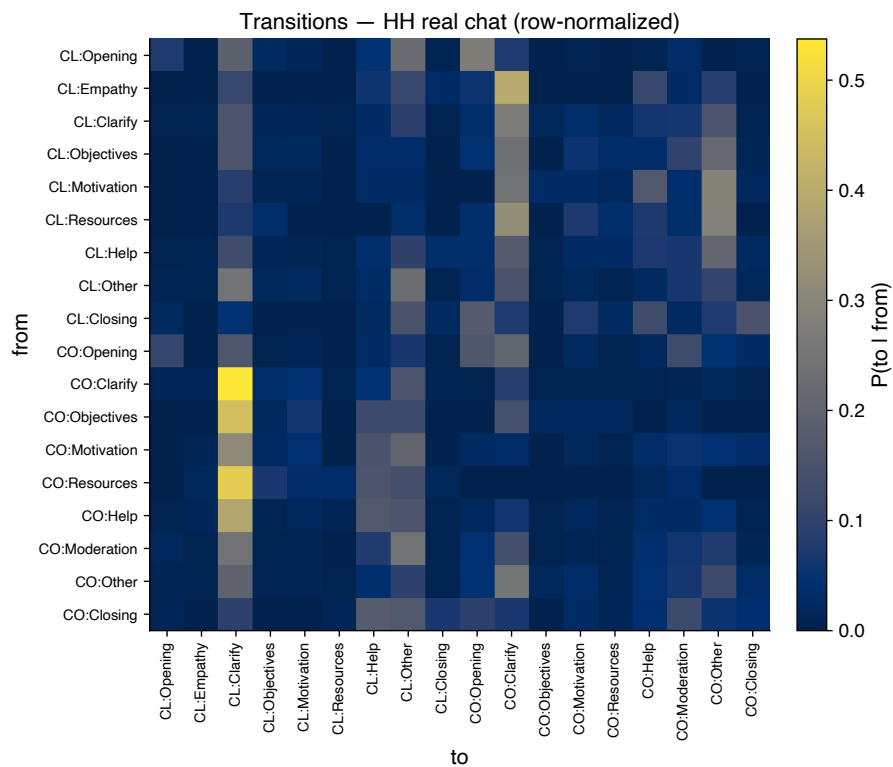


Figure S3. Transition heatmap for HH real (row-normalized).

Table S1. Personas used in chat roleplays (short descriptions truncated to the first sentence).

#	Persona name(s)	Conv.	Case brief (first sentence)
1	Luisa	89	Luisa ist genervt von den Streitereien ihrer Eltern.
2	Michael Schlosser	70	Herr Schlosser möchte gerne wieder eine Arbeit finden.
3	Elke	61	Elke macht sich Sorgen um ihren Sohn.
4	Jessica Bergmann	53	Jessica ist sich unsicher.
5	Marlene Grafe	43	Sie sucht eine besser bezahlte Stelle im Hauswirtschaftsbereich oder in der Gastronomie in Teilzeit (30 Stunden).
6	Marco	29	Marco hat Schwierigkeiten, sich selbst zu organisieren.
7	Eddie, Jakob, Lina, Susi, Tanja Müller	28	Lina befindet sich derzeit im Entzug.
8	Frau Wolke	25	Frau Wolkes Sohn Jan ist 14 Jahre alt und wohnt zuhause bei ihr.
9	Eddie	19	Eddie macht sich Sorgen um seinen Sohn.
10	Frau Schuster	13	Frau Schusters Sohn Max ist 8 Jahre alt geht zur Schule und ist oft anschließend im Hort.
11	Michaela	9	Verbesserung der Kommunikation innerhalb der Familie
12	Regina	6	Persona arbeitet 20 Jahre im gleichen Job und kann nicht mehr die Tätigkeit ausführen, hat körperliche Beschwerden, wenn sie nur an die Arbeit denkt; möchte\ldots
13	Lars	6	Lars befindet sich derzeit im Entzug.
14	Boris	5	Familienberatung suchen, um eine neue Perspektive auf familiäre Konflikte zu gewinnen.
15	Lara	4	Lara "funktioniert zwar nach außen hin gut", fühlt sich aber innerlich unter Druck und überfordert.
16	Mira	4	Mira fühlt sich zunehmend emotional erschöpft und überfordert im Studium.
17	Leyla	3	Leyla steht kurz nach dem Abitur und ist unsicher, welchen beruflichen Weg sie einschlagen möchte.
18	Annika	3	Annika kommt oft erst spät abends nach Hause in ihre 2-Zimmer-Wohnung in Berlin.
19	Michael Müller	2	Bewältigung der emotionalen Auswirkungen der Scheidung
20	Ute Meier	2	Unterstützung bei der Organisation eines stabilen Alltags nach der Scheidung.
21	Robert 12	2	Probleme des Soiihnen
22	Charlie	1	"Ich weiß nicht mehr weiter ... es macht einfach keinen Sinn mehr." Charlie hat ihr Mediendesign-Studium vor wenigen Monaten abgebrochen.
23	Elias	1	Elias hat das Gefühl, dass sein aktuelles Studium keine sichere Perspektive ist – er fühlt sich unter Druck sich gegen die Konkurrenz durchzusetzen und hat\ldots
24	Jonas	1	Jonas steht kurz vor dem Studienabschluss – er muss noch ein Seminar absolvieren und seine Bachelorarbeit schreiben.
25	Marie	1	Trauer
26	Lars	1	Unterstützung bei der Überwindung der Glücksspielsucht
27	Tobias Meier	1	Tobias sucht Unterstützung bei seiner Alkoholsucht.

Table S2. Persistence of top label shifts across conversations. Med. diff is the difference in *per-conversation median* label proportion, unweighted across conversations. It can diverge from the pooled shifts reported for RQ1 in the main paper (Section 5.1) when conversations contribute few spans. Conversations that never use a label contribute zero rather than being dropped. AUC is the more reliable, rank-based summary. It gives the probability that a random conversation from condition A exceeds one from condition B. $P(a>med)$ is the fraction of condition-A conversations above the condition-B median.

Comparison	Speaker	Label	Med. diff	AUC	$P(a>med)$
HH roleplay vs H-LLM roleplay	Client	Client: Consent	+0.186	0.925	0.971
HH roleplay vs H-LLM roleplay	Client	Client: Other statements	+0.103	0.837	0.882
HH roleplay vs H-LLM roleplay	Client	Client: Report on the implementation of recommendations for action	-0.045	0.249	0.059
HH roleplay vs H-LLM roleplay	Client	Client: Articulation of reasons for a change in the client	-0.037	0.298	0.103
HH roleplay vs H-LLM roleplay	Client	Client: Objective of the assignment	-0.026	0.346	0.294
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Moderation	+0.067	0.714	0.824
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Professional situation	+0.000	0.352	0.059
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Request for concerns	+0.014	0.632	0.647
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Complex Reflection	-0.024	0.419	0.368
HH roleplay vs H-LLM roleplay	Counselor	Counselor: Definition of counseling objectives	+0.000	0.435	0.485
HH real vs HH roleplay	Client	Client: Other statements	+0.112	0.880	0.981
HH real vs HH roleplay	Client	Client: Consent	-0.062	0.377	0.245
HH real vs HH roleplay	Client	Client: Objective of the assignment	-0.033	0.347	0.226
HH real vs HH roleplay	Client	Client: Formalities for conclusion	-0.034	0.249	0.094
HH real vs HH roleplay	Client	Client: Feedback on previous attempts at solutions	-0.018	0.401	0.264
HH real vs HH roleplay	Counselor	Counselor: Inappropriate remark	+0.039	0.757	0.717
HH real vs HH roleplay	Counselor	Counselor: Targeted, precise request	+0.054	0.755	0.774
HH real vs HH roleplay	Counselor	Counselor: Request for concerns	-0.009	0.419	0.377
HH real vs HH roleplay	Counselor	Counselor: Simple reflection	-0.016	0.443	0.415
HH real vs HH roleplay	Counselor	Counselor: Offer to use the counselor's professional resources	-0.031	0.295	0.075

Table S3. Full client-side OnCoCo label distribution (proportions) for the three chat conditions. The table lists labels reaching ≥ 0.01 in any condition.

Label	HH rp	HH real	H-LLM
Problem statement	0.083	0.087	0.109
Problem definition	0.097	0.085	0.093
Objective of the assignment	0.039	0.016	0.078
Formalities at the beginning and greeting	0.054	0.047	0.070
Report on the implementation of recommendations for action	0.010	0.012	0.057
Considering resource activation at a professional level	0.023	0.007	0.054
Articulation of reasons for a change in the client	0.012	0.025	0.052
General positive feedback	0.046	0.047	0.049
Positive feedback on specific recommendations for action	0.015	0.008	0.044
Consent	0.223	0.171	0.044
Feedback on previous attempts at solutions	0.042	0.023	0.043
Considering resource activation at the level of friends and family	0.009	0.004	0.043
Extension of the assignment	0.019	0.018	0.041
Disclosure of personal data	0.035	0.029	0.041
Other statements	0.108	0.241	0.029
Empathy for third parties or related to the present situation (PT)	0.006	0.006	0.020
Eliciting/Evoking "Change-Talk" (MI)	0.014	0.014	0.019
Final success	0.006	0.004	0.017
Own emotional expression	0.012	0.030	0.015
Further use of professional resources of counselor	0.018	0.004	0.015
Concern for another person (EC)	0.010	0.003	0.014
Formalities for conclusion	0.033	0.011	0.014
General request	0.040	0.027	0.012
Negative feedback on specific action recommendation	0.004	0.014	0.011
Inappropriate remark	0.013	0.021	0.006
Rejection	0.022	0.029	0.005

Table S4. Full counselor-side OnCoCo label distribution (proportions) for the three chat conditions. The table lists labels reaching ≥ 0.01 in any condition.

Label	HH rp	HH real	H-LLM
Other statements	0.146	0.145	0.146
Simple reflection	0.101	0.082	0.100
Formalities at the beginning/Opening of the conversation	0.102	0.088	0.089
Moderation	0.122	0.107	0.077
Complex Reflection	0.025	0.024	0.040
Request for concerns	0.060	0.039	0.036
Definition of counseling objectives	0.021	0.007	0.035
Thanks and appreciation to the client	0.036	0.021	0.034
Request for previous attempts at solutions	0.032	0.018	0.033
Targeted, precise request	0.025	0.067	0.032
Professional situation	0.002	0.005	0.030
Offer to use the counselor's professional resources	0.026	0.009	0.028
Professional, formal, institutional recommendation	0.016	0.005	0.028
Request for emotional state	0.030	0.036	0.026
Social relationships	0.021	0.019	0.026
Request for objectives/wishes	0.017	0.013	0.025
Farewell	0.031	0.022	0.021
Articulation of the perceived ability to change in the client (MI)	0.011	0.008	0.018
Request for personal data	0.023	0.037	0.016
Eliciting/Evoking "Change-Talk" (MI)	0.006	0.010	0.014
Advice	0.015	0.019	0.011
Health (mental and physical)	0.022	0.019	0.011
Evaluation, Interpretation	0.010	0.026	0.011
Leisure	0.005	0.009	0.010
Suggestion for resource activation at professional level	0.019	0.020	0.010
Inappropriate remark	0.011	0.065	0.008
Wish	0.010	0.014	0.008

Table S5. HMM state summaries (three states per condition, best of 20 random initialisations). States are ordered by occupancy (Occ.), the share of spans assigned to a state, so state 0 is the most frequent.

Condition	State	Occ.	Top labels (prob.)
HH roleplay	0	0.73	Clarify (0.57), Other (0.14), Moderation (0.07)
	1	0.19	Closing (0.25), Help (0.21), Other (0.17)
	2	0.08	Opening (0.69), Clarify (0.23), Other (0.04)
HH real	0	0.80	Clarify (0.53), Other (0.22), Help (0.07)
	1	0.15	Other (0.32), Help (0.20), Clarify (0.11)
	2	0.06	Opening (0.65), Other (0.14), Clarify (0.14)
H-LLM roleplay	0	0.50	Clarify (0.58), Other (0.09), Objectives (0.09)
	1	0.39	Help (0.27), Clarify (0.13), Other (0.12)
	2	0.10	Opening (0.54), Clarify (0.30), Objectives (0.04)

Table S6. HMM state-count sensitivity. Δ BIC is relative to each condition's best value (bold, $\Delta = 0$). "Optimal k " gives the state count minimising BIC and the one maximising 5-fold cross-validated held-out log-likelihood (conversation-level folds). Each cell is the best of 20 random initialisations. The cross-validated optimum grows with corpus size (4, 6 and 8 states). We therefore hold $k = 3$ fixed across conditions. The BIC optimum is not monotone in corpus size.

Condition	Spans	Δ BIC by state count k							Optimal k	
		2	3	4	5	6	7	8	BIC	CV
HH roleplay	3,927	469	70	0	53	187	262	438	4	4
HH real	4,927	302	0	12	112	260	403	502	3	6
H-LLM roleplay	21,117	5556	2694	1385	621	133	0	220	7	8

Table S7. Within-condition variability: conversation-level JSD to the condition mean.

Condition	Speaker	Median JSD	P90 JSD	Outlier frac
HH roleplay	Client	0.451	0.573	0.015
HH roleplay	Counselor	0.488	0.642	0.044
HH real	Client	0.384	0.458	0.038
HH real	Counselor	0.454	0.556	0.000
H-LLM roleplay	Client	0.542	0.620	0.012
H-LLM roleplay	Counselor	0.559	0.641	0.002

Table S8. RQ3 multiplicity control: all thirteen items of the simulated-client battery, comparing the two most recent semesters of the course line (GPT-OSS-120B, $n=23$, vs Llama 3.3 70B, $n=26$). Each row is a two-sided Mann-Whitney test with Cliff's δ as effect size. Means are on a 5-point disagreement-coded scale (1=agree, 5=disagree), so a negative δ means the GPT-OSS semester was rated more human-like. p_{Holm} is the Holm step-down adjustment across all thirteen items. Only the realism item (A270_02) survives ($p_{\text{Holm}}=.026$). The five acceptance constructs (perceived usefulness, ease of use, enjoyment, behavioral intention, social influence), tested the same way with Holm correction within the construct family, do not separate the semesters (all raw $p \geq .17$, max $|\delta|=0.23$). Rows are ordered by raw p .

Item	Content	$M_{\text{GPT-OSS}}$	M_{Llama}	U	p	p_{Holm}	δ
A270_02	Felt like writing with a real person	2.35	3.42	150	0.002	0.026	-0.50
A270_04	Individual messages were excellent	2.39	3.23	180	0.013	0.155	-0.40
A270_05	Responded understandably to my messages	2.39	2.96	208	0.057	0.632	-0.30
A270_07	Messages coherent and clearly phrased	2.22	2.58	226	0.118	1.000	-0.24
A270_12	No trouble understanding the messages	1.74	1.54	358	0.192	1.000	+0.20
A270_10	Satisfied with the interaction	2.39	2.73	238	0.201	1.000	-0.21
A270_01	Individual messages fit the conversation	2.04	2.35	248	0.259	1.000	-0.17
A270_03	Language quality of the messages	1.87	1.65	345	0.323	1.000	+0.15
A270_13	Reacted adequately to my follow-up questions	2.61	2.85	258	0.393	1.000	-0.14
A270_08	The conversation as a whole was coherent	2.52	2.78	228	0.413	1.000	-0.14
A270_06	Engaged with my questions and suggestions	2.74	2.85	273	0.581	1.000	-0.09
A270_11	Personality felt likeable	2.35	2.23	324	0.599	1.000	+0.09
A270_09	Could run a complete counseling conversation	2.57	2.65	296	0.959	1.000	-0.01

Table S9. Noise-band placement of the two chat comparisons under each unit of analysis, all referenced to HH real's own split-half band (1,000 splits). "Pctl." is the empirical percentile of the observed distance within that band. The pattern reported in the main text is unit-invariant: under every unit, and on both roles, both comparisons lie beyond the band.

Unit	Comparison	Role	JSD	Band mean	Band P95	Pctl.	V
SaT spans	HH roleplay	Client	0.233	0.109	0.133	100.0	0.266
	HH roleplay	Counselor	0.242	0.143	0.168	100.0	0.270
	H-LLM	Client	0.453	0.108	0.132	100.0	0.461
	H-LLM	Counselor	0.276	0.143	0.168	100.0	0.262
Regex spans	HH roleplay	Client	0.222	0.107	0.133	100.0	0.253
	HH roleplay	Counselor	0.236	0.154	0.176	100.0	0.268
	H-LLM	Client	0.342	0.107	0.132	100.0	0.291
	H-LLM	Counselor	0.258	0.154	0.178	100.0	0.227
Messages	HH roleplay	Client	0.242	0.138	0.170	100.0	0.276
	HH roleplay	Counselor	0.298	0.173	0.205	100.0	0.330
	H-LLM	Client	0.475	0.137	0.171	100.0	0.501
	H-LLM	Counselor	0.320	0.173	0.205	100.0	0.301

Table S10. The central RQ1 gap restricted to confidently labelled spans. Δ is the proximity gap of the main paper with a 95% interval from 20,000 conversation-level bootstrap resamples, and * marks intervals excluding zero. The "all items" row reproduces the SaT row of Table 3 in the main paper, interval included, and its bounds are taken from that run rather than redrawn here. If classifier error generated the effect, discarding the least confidently labelled items would reduce Δ . On the client side it does not, and Δ_{CL} increases. On the counselor side Δ_{CO} decreases and loses significance at the strictest threshold.

Confidence threshold	Role	Items retained	JSD real-LLM	JSD roleplay-LLM	Δ [95% CI]
all items	Client	100%	0.453	0.375	+0.078* [0.043, 0.107]
all items	Counselor	100%	0.276	0.196	+0.080* [0.042, 0.107]
≥ 0.70	Client	78%	0.466	0.385	+0.081* [0.041, 0.113]
≥ 0.70	Counselor	77%	0.264	0.211	+0.053* [0.015, 0.084]
≥ 0.90	Client	63%	0.479	0.384	+0.096* [0.056, 0.130]
≥ 0.90	Counselor	62%	0.253	0.210	+0.043* [0.004, 0.074]
≥ 0.99	Client	42%	0.455	0.377	+0.078* [0.031, 0.119]
≥ 0.99	Counselor	33%	0.265	0.242	+0.024 [-0.025, 0.063]



Figure S4. t-SNE projection of sentence embeddings by condition (Client).

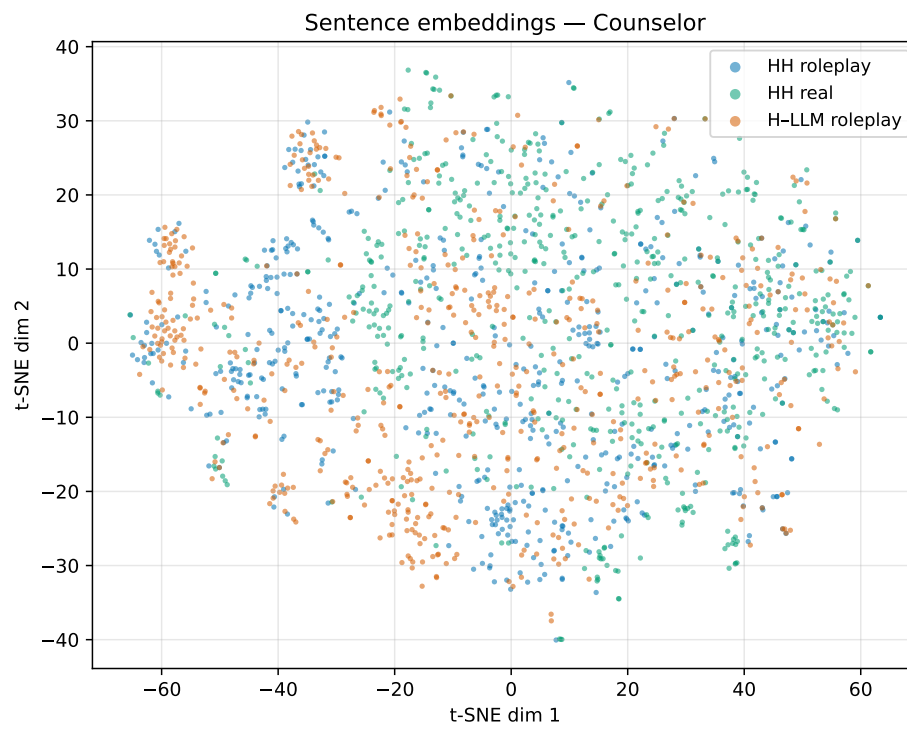


Figure S5. t-SNE projection of sentence embeddings by condition (Counselor).

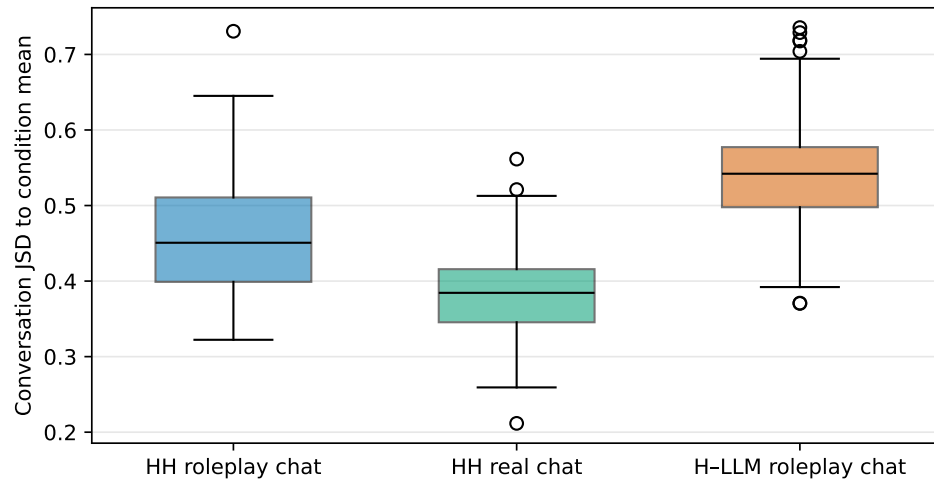


Figure S6. Within-condition variability (Client): distribution of conversation-level JSD to the condition mean.

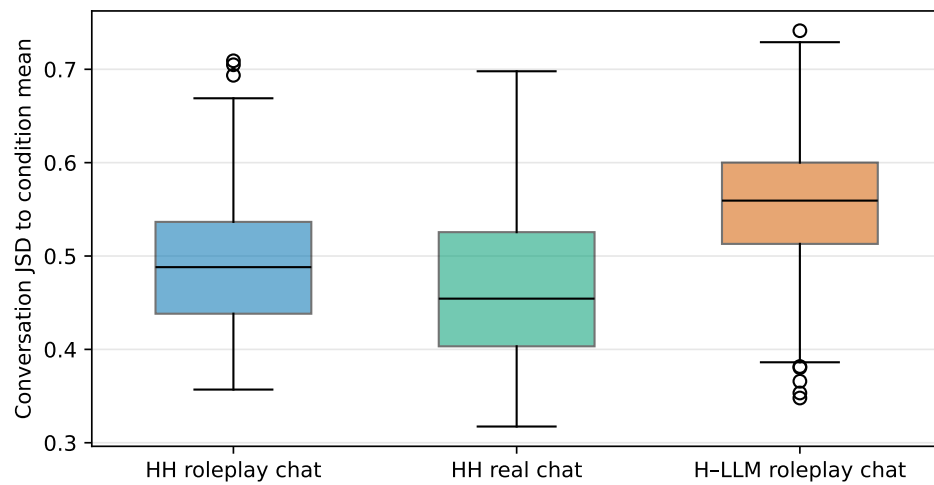


Figure S7. Within-condition variability (Counselor): distribution of conversation-level JSD to the condition mean.

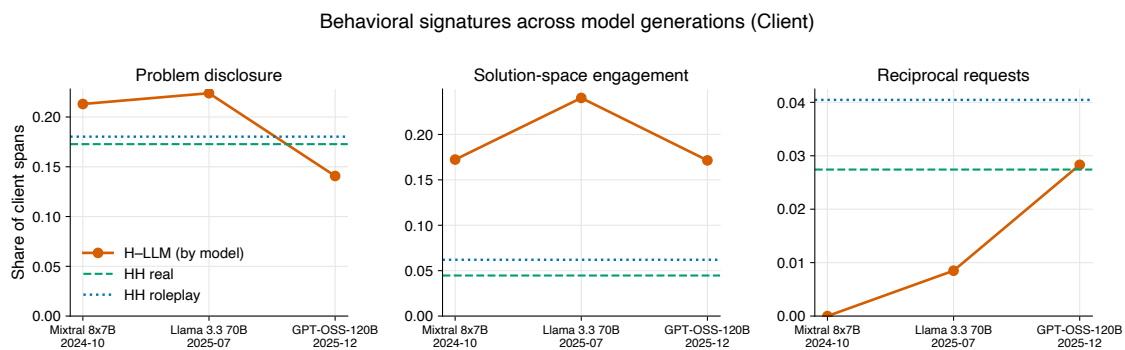


Figure S8. Behavioral signatures across model generations (share of Client spans), with human baselines (dashed for HH real, dotted for HH roleplay). Reciprocal questioning and restraint in problem disclosure improve for the newest model, but solution-space drift persists across generations.

Table S11. Between-condition distance and central gap under each exclusion set, with k the size of the label support after removal and renormalization. Band statistics use HH real’s own split-half band recomputed on the reduced support. Δ carries a 95% interval from 20,000 conversation-level bootstrap resamples, and * marks intervals excluding zero. The “full” row reproduces the SaT row of Table 3 in the main paper, interval included. Its bounds are taken from that run rather than redrawn here, so the baseline is one number and not two independent estimates of it.

Exclusion set	Role	k	JSD real-LLM	Band mean	Pctl.	Δ [95% CI]
full	Client	38	0.453	0.109	100.0	+0.078* [0.042, 0.109]
full	Counselor	57	0.276	0.144	100.0	+0.080* [0.042, 0.106]
S1	Client	33	0.460	0.108	100.0	+0.067* [0.028, 0.098]
S1	Counselor	51	0.243	0.149	100.0	+0.039* [0.001, 0.070]
S1+S2	Client	32	0.391	0.122	100.0	+0.008 [-0.029, 0.043]
S1+S2	Counselor	50	0.267	0.162	100.0	+0.043* [0.002, 0.077]
S1+S2+S3	Client	31	0.394	0.121	100.0	+0.011 [-0.026, 0.046]
S1+S2+S3	Counselor	48	0.270	0.161	100.0	+0.044* [0.002, 0.077]

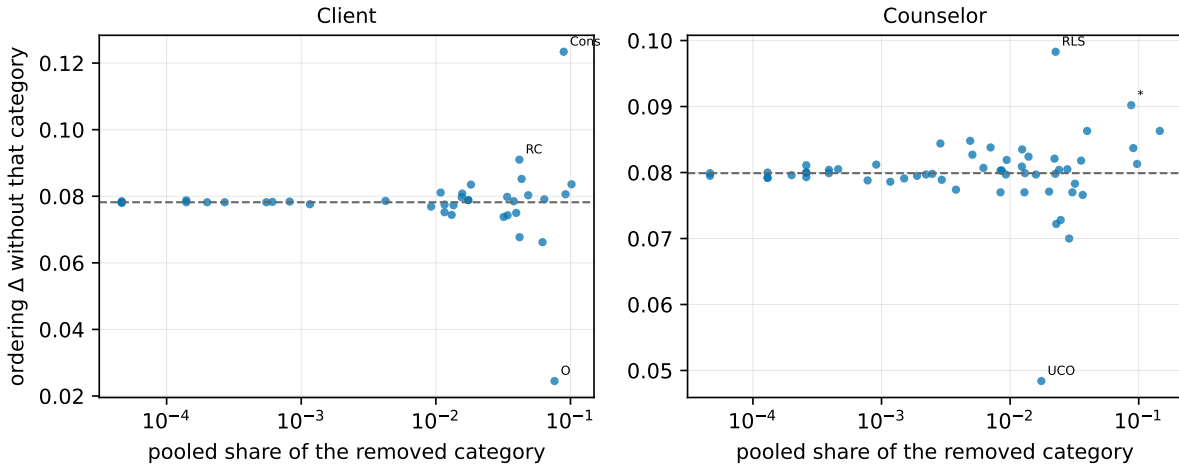


Figure S9. Leave-one-out sweep: the proximity gap Δ recomputed with each category removed in turn, plotted against that category’s pooled share. The dashed line is Δ on the full support. No single removal drives Δ to zero on either role, and the client-side category whose removal moves Δ most (*Consent*) moves it upward rather than downward.

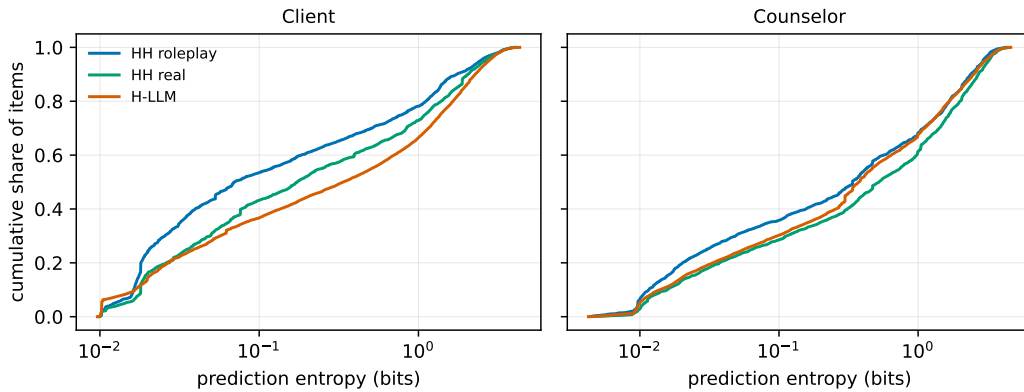


Figure S10. Cumulative distribution of prediction entropy by condition and role (symlog x -axis). HH real carries the heaviest uncertainty tail on both roles. The pooled client-side confidence ranking in the main text is driven by span length, which the length-matched comparison controls.

Table S12. Numbers behind Figure 3 of the main paper: between-condition JSD anchored in the reference condition’s split-half noise band (1,000 splits), with band mean (noise floor), band P95 (upper edge), the 95th percentile of the size-matched null (nm-P95), and Cramér’s V . All comparisons use HH real as reference.

Comparison	Role	JSD	Band mean	Band P95	nm-P95	V
HH roleplay	Client	0.233	0.109	0.133	0.092	0.27
HH roleplay	Counselor	0.242	0.143	0.168	0.116	0.27
H-LLM	Client	0.453	0.108	0.132	0.073	0.46
H-LLM	Counselor	0.276	0.143	0.168	0.091	0.26
H-LLM [GPT-OSS-120B]	Client	0.361	0.108	0.131	0.087	0.41
H-LLM [GPT-OSS-120B]	Counselor	0.293	0.143	0.167	0.111	0.33
H-LLM [Llama 3.3 70B]	Client	0.524	0.109	0.132	0.076	0.57
H-LLM [Llama 3.3 70B]	Counselor	0.298	0.144	0.169	0.098	0.30
H-LLM [Mixtral 8x7B]	Client	0.495	0.109	0.132	0.086	0.54
H-LLM [Mixtral 8x7B]	Counselor	0.292	0.144	0.170	0.110	0.33

Table S13. Distributional proximity to HH real by model with 95% conversation-level bootstrap CIs (lower = closer). These are the intervals drawn in the per-model rows of Figure 3 of the main paper.

Model	Client JSD [95% CI]	Counselor JSD [95% CI]
GPT-OSS-120B	0.361 [0.338, 0.402]	0.293 [0.274, 0.338]
Llama 3.3 70B	0.524 [0.503, 0.552]	0.298 [0.286, 0.332]
Mixtral 8x7B	0.495 [0.468, 0.530]	0.292 [0.272, 0.338]

Table S14. Sequential structure summary (role-aware transitions) for the three chat conditions. All three are dominated by cross-role alternation. Section 5.3 of the main paper reads these differences.

Condition	Same-role	Cross-role	Self-transition
HH roleplay	0.323	0.677	0.101
HH real	0.365	0.635	0.115
H-LLM roleplay	0.292	0.708	0.070

Table S15. All 70 transition cells that shift significantly for H-LLM vs HH real under FDR control at $q = 0.10$, sorted by absolute shift and set in two column blocks. Rates are row-normalized transition probabilities in each condition. Δ is the H-LLM rate minus the HH real rate, n the combined transition count, and p the empirical p -value from 2,000 split-half resamples of HH real. The smallest attainable value is shown as <0.001 . Section 5.3 of the main paper reports the counts and the largest shifts.

From	To	H-LLM	HH real	Δ	n	p	From	To	H-LLM	HH real	Δ	n	p
CL:Opening	CL:Clarify	0.585	0.192	+0.394	441	<0.001	CL:Clarify	CO:Opening	0.100	0.036	+0.064	421	<0.001
CO:Closing	CL:Help	0.480	0.178	+0.302	260	0.002	CO:Clarify	CL:Resources	0.070	0.008	+0.062	292	<0.001
CO:Resources	CL:Resources	0.337	0.035	+0.302	97	<0.001	CL:Help	CO:Motivation	0.087	0.029	+0.058	156	0.004
CL:Opening	CO:Opening	0.004	0.275	-0.271	36	<0.001	CL:Other	CO:Other	0.048	0.106	-0.058	84	0.011
CO:Objectives	CL:Resources	0.221	0.000	+0.221	146	<0.001	CO:Other	CO:Objectives	0.075	0.020	+0.055	136	<0.001
CO:Moderation	CL:Other	0.025	0.243	-0.218	85	<0.001	CL:Opening	CL:Empathy	0.055	0.000	+0.055	39	<0.001
CO:Help	CL:Clarify	0.175	0.384	-0.209	214	<0.001	CO:Help	CL:Motivation	0.072	0.020	+0.052	61	0.004
CO:Objectives	CL:Objectives	0.227	0.020	+0.206	151	<0.001	CO:Other	CO:Help	0.091	0.042	+0.049	174	0.003
CO:Motivation	CL:Other	0.009	0.203	-0.194	31	0.021	CO:Other	CO:Motivation	0.079	0.032	+0.047	149	0.002
CL:Opening	CL:Other	0.052	0.225	-0.173	64	0.018	CO:Clarify	CL:Motivation	0.081	0.044	+0.037	368	0.015
CO:Other	CL:Clarify	0.031	0.197	-0.165	151	<0.001	CO:Other	CO:Closing	0.068	0.032	+0.036	131	0.004
CO:Help	CL:Help	0.318	0.167	+0.151	284	0.001	CL:Other	CL:Closing	0.040	0.006	+0.034	18	<0.001
CO:Help	CL:Other	0.017	0.157	-0.140	44	0.013	CO:Clarify	CL:Empathy	0.045	0.013	+0.033	196	<0.001
CL:Closing	CL:Help	0.153	0.025	+0.129	36	<0.001	CO:Moderation	CL:Closing	0.036	0.004	+0.032	31	<0.001
CO:Motivation	CL:Resources	0.127	0.000	+0.127	112	<0.001	CL:Other	CL:Motivation	0.054	0.024	+0.030	34	0.012
CO:Help	CL:Resources	0.134	0.010	+0.124	108	<0.001	CL:Resources	CO:Closing	0.030	0.000	+0.030	29	<0.001
CO:Clarify	CL:Other	0.033	0.153	-0.120	266	<0.001	CO:Moderation	CO:Objectives	0.033	0.004	+0.029	29	<0.001
CO:Motivation	CO:Clarify	0.152	0.035	+0.117	138	<0.001	CL:Empathy	CO:Objectives	0.028	0.000	+0.028	10	<0.001
CL:Clarify	CO:Clarify	0.379	0.273	+0.106	1776	0.009	CO:Objectives	CL:Closing	0.027	0.000	+0.027	18	<0.001
CO:Moderation	CL:Objectives	0.113	0.008	+0.105	97	<0.001	CL:Other	CL:Resources	0.031	0.005	+0.027	14	<0.001
CL:Opening	CL:Help	0.151	0.050	+0.101	114	0.008	CL:Help	CO:Objectives	0.035	0.008	+0.026	61	<0.001
CO:Moderation	CL:Help	0.174	0.076	+0.098	166	0.002	CO:Closing	CL:Objectives	0.025	0.000	+0.025	13	<0.001
CO:Other	CL:Other	0.001	0.096	-0.095	50	0.003	CO:Closing	CL:Motivation	0.025	0.000	+0.025	13	<0.001
CL:Objectives	CO:Objectives	0.095	0.000	+0.095	115	<0.001	CO:Moderation	CL:Motivation	0.036	0.011	+0.024	33	0.008
CO:Help	CL:Objectives	0.101	0.010	+0.091	82	<0.001	CO:Opening	CO:Objectives	0.023	0.000	+0.023	22	<0.001
CL:Help	CL:Other	0.012	0.100	-0.088	45	0.024	CO:Other	CO:Resources	0.031	0.010	+0.021	58	0.018
CL:Closing	CO:Moderation	0.110	0.025	+0.085	26	<0.001	CL:Opening	CL:Resources	0.021	0.000	+0.021	15	<0.001
CL:Resources	CO:Objectives	0.083	0.000	+0.083	81	<0.001	CL:Clarify	CL:Objectives	0.034	0.015	+0.019	147	0.019
CL:Clarify	CL:Other	0.013	0.093	-0.079	168	<0.001	CL:Other	CL:Opening	0.029	0.011	+0.017	17	0.028
CL:Motivation	CO:Motivation	0.108	0.029	+0.079	78	0.002	CO:Help	CL:Closing	0.021	0.005	+0.017	18	<0.001
CL:Other	CL:Help	0.105	0.030	+0.075	56	<0.001	CO:Opening	CL:Empathy	0.015	0.000	+0.015	15	<0.001
CO:Clarify	CL:Objectives	0.108	0.037	+0.071	470	<0.001	CL:Help	CL:Resources	0.019	0.004	+0.015	33	<0.001
CL:Help	CO:Closing	0.091	0.025	+0.066	162	<0.001	CO:Motivation	CO:Objectives	0.013	0.000	+0.013	11	<0.001
CL:Other	CL:Objectives	0.083	0.018	+0.065	40	<0.001	CL:Clarify	CL:Resources	0.020	0.008	+0.012	85	0.012
CO:Moderation	CL:Resources	0.064	0.000	+0.064	54	<0.001	CL:Resources	CL:Help	0.010	0.000	+0.010	10	<0.001

Table S16. All 20 transition cells that shift significantly for HH roleplay vs HH real under FDR control at $q = 0.10$, sorted by absolute shift. Columns as in Table S15.

From	To	HH roleplay	HH real	Δ	n	p
CL:Opening	CL:Clarify	0.422	0.192	+0.230	66	<0.001
CO:Closing	CL:Closing	0.287	0.069	+0.219	32	<0.001
CO:Motivation	CO:Clarify	0.232	0.035	+0.196	29	<0.001
CO:Moderation	CL:Other	0.056	0.243	-0.187	78	0.001
CO:Moderation	CL:Clarify	0.422	0.243	+0.178	169	<0.001
CO:Other	CL:Clarify	0.051	0.197	-0.146	114	0.001
CL:Resources	CO:Moderation	0.161	0.036	+0.126	11	0.001
CO:Resources	CL:Resources	0.122	0.035	+0.087	11	0.011
CL:Help	CO:Closing	0.103	0.025	+0.078	22	<0.001
CO:Moderation	CL:Closing	0.072	0.004	+0.069	19	<0.001
CO:Help	CL:Empathy	0.070	0.015	+0.054	11	<0.001
CO:Other	CO:Help	0.085	0.042	+0.043	48	0.007
CO:Other	CO:Objectives	0.060	0.020	+0.040	29	0.003
CL:Clarify	CL:Other	0.054	0.093	-0.039	175	0.016
CO:Moderation	CL:Objectives	0.044	0.008	+0.037	13	<0.001
CL:Other	CL:Objectives	0.049	0.018	+0.032	22	0.001
CO:Other	CO:Closing	0.063	0.032	+0.031	36	0.013
CO:Other	CO:Resources	0.038	0.010	+0.028	17	<0.001
CO:Clarify	CL:Resources	0.025	0.008	+0.017	25	0.009
CL:Clarify	CO:Closing	0.021	0.008	+0.013	32	0.008